	Sun 07/13	Mon 07/14	Tue 07/15	Wed 07/16	Thu 07/17	Fri 07/18	Sat 07/19	Sun 07/20
8:00		Arrival	Arrival	Arrival	Arrival	Arrival		
8:30		Basic Tools	Optics 2	Optics 4 [2]	Optics 5 [3]	Optics 6 [4]		
9:00		[12]	Imaging System [13]	Telescopes	Spectroscopy	Detecting Exoplanets		
9:30		Break	Break	Break	Break	Break		
9:45		Basic Tools	IC/DAQ2 [15]	Electronics 3 [16]	Electronics 5 [17]	Electronics 7 [18]		
10:15		[27]	Basic Python					
10:45		Break	Break	Break	Break	Break		
11:00		Optics 1 [29]	IC/DAQ3 [30]	IC/DAQ 4 [31]	IC/DAQ 5 [32]	IC/DAQ 6 [33]		
11:30		Photons, Waves, & Fluxes	Problem Set [42]	Fourier transform and properties	Sampling + aliasing	Problem set		
12:00	Arrival							
12:30		Lunch	Lunch	Lunch	Lunch	Lunch		
1:00							Personal Time	Personal Time
1:30		Particle Detection 1 [44]	Optics 3 [45]	Radio 1 [46]	Radio 3 [47]	Radio 4 [48]		
2:00			Imaging System			RTL-SDR/data processing		
2:30		Break	Break	Break	Break	Break		
2:45		Electronics 1 [59]	Electronics 2 [60]	Electronics 4 [61]	Particle Detection 2 [62	Particle Detection 3 [63]		
3:15				[71]	[72]			
3:45		Break	Break	Break	Break	Break		
4:00		IC/DAQ 1 [74]	IC/DAQ 3 [75]	Radio 2 [76]	Electronics 6 [77]	Electronics 8 [78]		
4:30		Setup/data types	Problem set	[87]				
5:00		Personal Time	Personal Time	Personal Time	Personal Time	Personal Time		
5:30								
6		Dinner	Dinner	Dinner	Dinner	Dinner		
7								
8								
9								
10								
11								
12								

	Mon 07/21	Tue 07/22	Wed 07/23 [1]			Thus 07/24			Fri 07/25	Sat 07/26
8:00	Arrival	Arrival	Arrival			Arrival			Arrival	
8:30	Radio 5	Radio 6 [5]	IC/DAQ [6]	Cosmic Rays [7]	Radio Astro [8]	IC/DAQ [9]	Cosmic Rays [10]	Radio Astro [11]	Extra lab time & prep	
9:00			Project Lab Hr #1	Project Lab Hr #1	Project Lab Hr #1 [14]	Project Lab Hr #7	Project Lab Hr #7	Project Lab Hr #7		
9:30	Break	Break	Break			Break			Break	
9:45	Electronics 9 [19]	Electronics 11 [20]	IC/DAQ [21]	Cosmic Rays [22]	Radio Astro [23]	IC/DAQ [24]	Cosmic Rays [25]	Radio Astro [26]	Extra lab time & prep	
10:15			Project Lab Hr #2	Project Lab Hr #2	Project Lab Hr #2 [28]	Project Lab Hr #8	Project Lab Hr #8	Project Lab Hr #8		
10:45	Break	Break	Break			Break			Break	
11:00	Particle Detection 4 [34]	Particle Detection 6 [35]	IC/DAQ [36]	Cosmic Rays [37]	Radio Astro [38]	IC/DAQ [39]	Cosmic Rays [40]	Radio Astro [41]	Presentation	
11:30			Project Lab Hr #3	Project Lab Hr #3	Project Lab Hr #3 [43]	Project Lab Hr #9	Project Lab Hr #9	Project Lab Hr #9		
12:00										Departure
12:30	Lunch	Lunch	Lunch			Lunch			Lunch	
1:00										
1:30	Particle Detection 5 [49]	IC/DAQ 8 [50]	IC/DAQ [51]	Cosmic Rays [52]	Radio Astro [53]	IC/DAQ [54]	Cosmic Rays [55]	Radio Astro [56]	Presentation	
2:00	[57]		Project Lab Hr #4	Project Lab Hr #4	Project Lab Hr #4 [58]	Project Lab Hr #10	Project Lab Hr #10	Project Lab Hr #10		
2:30	Break	Break	Break			Break			Break	
2:45	IC/DAQ 7 [64]	IC/DAQ 9	IC/DAQ [65]	Cosmic Rays [66]	Radio Astro [67]	IC/DAQ [68]	Cosmic Rays [69]	Radio Astro [70]	Presentation	
3:15	Connecting to Hw, File I/O		Project Lab Hr #5	Project Lab Hr #5	Project Lab Hr #5 [73]	Project Lab Hr #11	Project Lab Hr #11	Project Lab Hr #11		
3:45	Break	Break	Break			Break			Break	
4:00	Electronics 10 [79]	Electronics 12 [80]	IC/DAQ [81]	Cosmic Rays [82]	Radio Astro [83]	IC/DAQ [84]	Cosmic Rays [85]	Radio Astro [86]	Presentation	
4:30	[88]		Project Lab Hr #6	Project Lab Hr #6	Project Lab Hr #6 [89]	Project Lab Hr #12	Project Lab Hr #12	Project Lab Hr #12		
5:00	Personal Time	Personal Time	Personal Time			Personal Time			Personal Time	
5:30										
6	Dinner	Dinner	Dinner			Dinner			Dinner	
7										
8										
9										
10		QNO telescope Observation (if weather permits)								
11										
12										

Radio 1 & 2	Pham Viet Dung	Antenna Basics, Design, and Simulation		**2 hr**	(14/7 26/7)
Radio 3	Phan Thanh Hien	Radio Astronomy Telescope: Front End Electroni		1 hr	(6/7 - 26/7)
Radio 4	Vo Bich Hien	Low Frequency Astronomical Array System		1hr	(21/7 26/7)
Radio 5 & 6	Han Huy Dung	Data Processing & FPGA		2 hr	(14/7 - 24/7)
	Hoang Kim Dinh	Python & Data Analysis			(14/7 26/7)
	Nguyen Luong Quang			21/7 - 26/7	
	LAB: Fabrication				
	LAB: Assemble Interferometry Array & Data Acquisiton			V Hien: buy SDR & electroncis from local stores,	
	Software	CST Student Version	Pham Viet Dung		

[1] Each of the three labs can take about 6 students for 2-3 groups setup

[2] Gene Serabyn

Telescope

[3] Gene Serabyn

[4] Gene Serabyn

[5] Hien Nguyen

[6] Logan & Sam

[7] Son Cao

[8] Bang Nhan
(session RA-LAB-00)
- Can take 2 groups, or 6 students

[9] Parallel lab (2 groups per project)

[10] Son Cao

[11] Bang Nhan

[12] Hien Nguyen
0. logistics
1. where, how bright and what color?
2. how to do these measurement?

Hien Nguyen
3. Understanding data
- Statistical Analysis
4. Fourier Series/Transform

[13] Jack Sayers
Galaxy Clusters - Cosmology, Astrophysics, and the Need for Multi-Probe Observations

[14] - Group teaming up & Lab setup
- Installing software for those struggled

[15] Sam&Chris
Python functions and control flow
problem sets available

[16] Silvia

[17] Silvia

[18] Silvia

[19] Hien Nguyen

[20] Hien Nguyen

[21] Logan & Sam

[22] Son Cao

[23] Bang Nhan
(session RA-LAB-01a)

[24] Parallel lab (3 groups each)
- IC/DAQ - groups with odd number
- RA - groups with even number

[25] Son Cao

[26] Bang Nhan

[27] Gene Serabyn
5. Writing proposal - Examples
a. FTS for CSO
b. MIRI for JWST
Hien Nguyen
6. Presenting your work - how to give a talk?

[28] Galactic 21-cm lab
- Astropy & Healpy exercises

[29] Gene Serabyn

[30] Logan&Sam. Problem sets on basic python of multiple difficulties available

[31] Sam Condon
Lecture on fourier series, fourier transform, and fourier transform properties

[32] Sam
Lecture on sampling, aliasing, and if time permits, the discrete fourier transform

[33] Logan and Sam
Set up file I/O for IC/DAQ lab next week
Set up measurement/analysis procedures for next week

[34] Michael Holik

[35] Michael Holik

[36] Logan & Sam

[37] Son Cao

[38] Bang Nhan
(session RA-LAB-01b)

[39] Parallel lab (3 groups each)
- IC/DAQ - groups with odd number
- RA - groups with even number

[40] Son Cao

[41] Bang Nhan

[42] - Radio astronomy introduction

[43] Galactic 21-cm lab
- Astropy & Healpy exercises

[44] Son Cao

[45] Gene Serabyn

[46] Pham Viet Dung
Antenna Simulation

[47] Phan Thanh Hien

[48] Hàn Huy Dũng

[49] Michael Holik

[50] Mikio

[51] Logan & Sam

[52] Son Cao

[53] Bang Nhan
(session RA-LAB-02a)

[54] Parallel lab (3 groups each)
- IC/DAQ - groups with odd number
- RA - groups with even number

[55] Son Cao

[56] Bang Nhan

[57] - Low-noise amplifier
- RF circuit components
- Component datasheet

[58] RF & VNA lab
- Measuring antenna & RF components

[59] Silvia Zhang

[60] Silvia

[61] Silvia

Laboratory

[62] Hoang Kim Dinh - Berkeley

[63] Hoang Kim Dinh - Berkeley

[64] Sam & Logan

[65] Logan & Sam

[66] Son Cao

[67] Bang Nhan
(session RA-LAB-02b)

[68] Parallel lab (3 groups each)
- IC/DAQ - groups with odd number
- RA - groups with even number

[69] Son Cao

[70] Bang Nhan

[71] - Basic Particle Detection

[72] - Antenna & transmission line
- Low-noise amplifier (LNA)

[73] SDR & GNU Radio labs

[74] Logan and Sam
Set up python/anaconda
Basic data types in python
problem sets available for those who finish early

[75] Sam and Chris.
Problem sets on basic python of multiple difficulties available

[76] Pham Viet Dung
Antenna Simulation - Cont.

[77] Tran Quang Vinh
Demonstration of Statistical Analysis.
https://drive.google.com/file/d/1KU3cCXnZq7xmH8jvvyOos_mx0CQG_BaF/view?usp=sharing

[78] Le Ngan

[79] Bang Nhan
(Session RA-03b)
- Radio observation simulation
- RF Nano VNA measurement
- SDR & GNU Radio intro

[80] Sam & Logan

[81] Logan & Sam

[82] Son Cao

[83] Bang Nhan
(session RA-LAB-03)

[84] Parallel lab (3 groups each)
- IC/DAQ - groups with odd number
- RA - groups with even number

[85] Son Cao

[86] Bang Nhan

[87] - Interferometry intro
- CASA Reduction pipeline introduciton for ALMA (followed up by Fri 7/26 for Photometry session)

[88] - Software Defined Radio (SDR) intro
- GNU Radio intro
- Basic FFT/spectrometer with GNU radio
- Digital Signal Processing (DSP)

[89] Extra lab time & presentation discussion